Michael Evans

mevansci.github.io

mevan028@odu.edu (757) 236-1257

EDUCATION

Old Dominion University Norfolk, VA
M.S. in Electrical & Computer Engineering Expected May 2027
Thesis: *AI Framework for Disease-Agnostic Brain Glucose Hypometabolism Prediction*
GPA: 3.68/4.00

B.S. in Computer Science May 2025
GPA: 3.90/4.00

Tidewater Community College Norfolk, VA
A.S. in Computer Science May 2023

RESEARCH INTERESTS

Deep learning, medical image analysis, neuroimaging, multimodal learning, neurodegenerative disease modeling

RESEARCH EXPERIENCE

Graduate Research Assistant Sept 2025-Present
Old Dominion University, advised by Dr. Khan Iftekharuddin
Vision Lab

- Trained a multimodal CNN-Transformer model from scratch to predict regional age-adjusted FDG-PET uptake, enabling hypometabolic profile prediction from MRI scans and CSF biomarkers.
- Developed CoMNet, a 3D MedNeXt ensemble with corrective diffusion framework that improved average Dice by up to 0.0611 and reduced between-fold prediction variability across three multi-site glioma datasets.
- Designed a standardized, BIDS-compliant preprocessing pipeline for multimodal neuroimaging (sMRI, fMRI, FDG-PET) using MATLAB, fMRIPrep, and Clinica, ensuring reproducible research outcomes.

Undergraduate Research Assistant Sept 2024-Sept 2025
Old Dominion University, advised by Dr. Khan Iftekharuddin
Vision Lab

- Trained a random forest classifier for atypical Alzheimer's disease classification using cognitive test scores and MRI features, improving recall from 34% to 77% on ADNI and 52% to 69% on NACC over baseline.
- Constructed a FreeSurfer pipeline on an HPC cluster to extract regional volumetric MRI features at scale.
- Visualized an ROI-map on an anatomically averaged brain template to highlight brain regions with significant features ($\alpha = 0.05$) for Alzheimer's classification with volumetric group mean z-score color coding.

NSF REU Intern - Developing Self-Drive Algorithms May-July 2024
Lawrence Tech and Michigan State University, advised by Drs. Chan-Jin Chung and Josh Siegel
Computer Science & Artificial Intelligence Robotics Lab

- Designed an adaptive speed algorithm to reduce acceleration and braking in autonomous vehicles through intersections by up to 75.35%, minimizing fuel consumption and noise pollution.
- Implemented and analyzed DBSCAN and K-means machine learning self-drive algorithms for reliability.
- Built a simulation in GazelleSim for testing our autonomous intersection control in a controlled environment.

NSF REU Intern - Disinformation Detection and Analytics June-Aug 2023
Old Dominion University, advised by Dr. Jian Wu
Lab for Applied Machine Learning and Natural Language Processing Systems

- Engineered prompts and tuned hyperparameters with the OpenAI API for scientific claim verification.
- Increased the size of our project corpus by over 200% to improve benchmarking the effectiveness of the GPT-3.5-turbo model's ability to generalize across multiple scientific news domains.
- Calculated the precision, recall, and F1 score on zero-shot classification with the GPT-3.5-turbo model against our dataset on two sub-tasks: stance labeling and identifying sentence rationales.

PUBLICATIONS

Preprints and Manuscripts Under Review

M. L. Evans, M. A. Witherow, M. F. B. Hossen, and K. M. Iftekharuddin. Transformer-Based Prediction of Regional Hypometabolism in Alzheimer's Disease From MRI and CSF Biomarkers.

M. L. Evans, M. F. B. Hossen, M. S. Sadique, W. Farzana, and K. M. Iftekharuddin. CoMNet: A MedNeXt-CorrDiff Framework for Multi-Site Brain Tumor Segmentation. *arXiv preprint*, 2026.

M. A. Witherow, M. L. Evans, A. Temtam, H. R. Okhravi, and K. M. Iftekharuddin. Machine Learning-Enhanced Non-Amnestic Alzheimer's Disease Diagnosis From MRI and Clinical Features. *arXiv preprint*, 2026. *Under review at the IEEE Journal of Biomedical and Health Informatics (JBHI).*

Peer-Reviewed Publications

M. L. Evans, M. Machado, R. Johnson, L. Escamilla, A. Vadella, B. Froemming-Aldanondo, T. Rastoskueva, M. Jostes, D. Butani, R. Kaddis, C. Chung, and J. Siegel. A Roadside Unit for Infrastructure Assisted Intersection Control of Autonomous Vehicles. *IEEE International Conference on Electro/Information Technology (EIT)*, 2025.

B. Froemming-Aldanondo, T. Rastoskueva, M. L. Evans, M. Machado, A. Vadella, L. Escamilla, R. Johnson, M. Jostes, D. Butani, R. Kaddis, C. Chung, and J. Siegel. Evaluating Low-Resource Lane Following Algorithms for Compute-Constrained Automated Vehicles. *IEEE International Conference on Artificial Intelligence, Robotics, and Control (AIRC)*, 2025.

M. L. Evans, D. Soós, E. Landers, and J. Wu. MSVEC: A Multidomain Testing Dataset for Scientific Claim Verification. *National Workshop for REU Research in Networking and Systems (REUNS @ MobiHoc)*, 2023.

SELECTED PROJECTS

Resting State Functional MRI Based Alzheimer's Disease Classification
ECE 784: Computer Vision Term Project

- Trained an SVM with SelectKBest feature selection to classify Alzheimer's disease from resting-state fMRI data on an ADNI cohort of 66 patients, achieving up to 0.71 balanced accuracy.
- Performed rs-fMRI preprocessing and AAL2 brain parcellation to extract: functional connectivity, amplitude of low-frequency fluctuations (ALFF), regional homogeneity (ReHo), and mean BOLD signal features.

Pre-Processing and Noise Distribution Learning in MRI
ECE 783: Digital Image Processing Term Project

- Applied non-local means filtering to denoise MRI corrupted by additive Rician noise, achieving a 39.8% SNR increase and experimentally verifying that the Rician distribution approaches Gaussian at high SNR.
- Modeled MRI acquisition noise by generating Rician samples from a bivariate normal distribution, and performed residual distribution analysis between the clean reference image and denoised images.

SELECTED TECHNOLOGIES

Programming Languages
Python, R, Bash, C++, MATLAB

Libraries & Frameworks
Medical Imaging: fMRIPrep, Clinica, FreeSurfer, SPM, MONAI, NiBabel, Nilearn
Machine Learning / General: PyTorch, OpenCV, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn

AWARDS

NSF CISE REU Travel Grant – $1,200	May 2025
Research Profile Highlight – Old Dominion University College of Sciences Newsletter	Nov 2024
BIO, GEO, CISE and OCE REU Travel Grant – $1,200	Sept 2023